

UST-ILIMSK, Russian Federation

WMO: 301170

Lat: 58.030N

Lon: 102.720E

Elev: 401

StdP: 96.60

Time Zone: 8.00 (E08)

Period: 90-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -38.1	(c) -35.1	(d) -41.3	(e) 0.1	(f) -37.8	(g) -37.5	(h) 0.1	(i) -34.6	(j) 10.0	(k) -9.7	(l) 9.1	(m) -13.0	(n) 2.1	(o) 210	(p) 0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	28.9	17.3	27.0	16.7	25.2	16.1	19.1	24.8	18.1	24.2	17.3	23.4	3.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.4	13.1	20.9	16.1	12.0	20.1	15.0	11.1	19.4	56.6	24.6	53.1	24.3	50.3	23.5	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.0	(b) 7.9	(c) 7.1	(e) -38.1	(f) 31.9	(g) 3.0	(h) 1.0	(i) -40.3	(j) 32.6	(k) -42.0	(l) 33.1	(m) -43.7	(n) 33.7	(o) -45.8	(p) 34.3		
(a) 9.0	(b) 7.9	(c) 7.1	(e) -38.5	(f) 20.8	(g) 2.9	(h) 1.3	(i) -40.6	(j) 21.7	(k) -42.3	(l) 22.5	(m) -43.9	(n) 23.2	(o) -46.0	(p) 24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.1	-20.8	-15.8	-8.9	-0.6	8.0	14.6	19.3	16.0	6.6	-0.9	-13.2	-18.5		
	DBStd	14.90	8.40	6.61	6.68	4.64	5.69	4.74	3.71	3.78	4.27	5.66	8.71	7.46		
	HDD10.0	4747	956	723	584	317	112	11	0	3	118	339	697	885		
	HDD18.3	7204	1214	957	843	567	325	133	34	91	353	597	947	1143		
	CDD10.0	691	0	0	0	1	48	148	289	189	16	1	0	0		
	CDD18.3	107	0	0	0	0	3	19	66	18	0	0	0	0		
	CDH23.3	838	0	0	0	0	38	159	532	109	0	0	0	0		
	CDH26.7	191	0	0	0	0	7	30	143	11	0	0	0	0		
	WSAvg	3.6	3.6	3.6	3.7	3.9	4.0	3.3	2.7	2.9	3.1	4.2	3.9	3.9		
	PrecAvg	364	17	13	13	21	32	45	59	53	38	28	26	23		
	PrecMax	479	37	24	28	52	65	96	130	113	100	55	49	51		
(15) Precipitation	PrecMin	292	7	3	1	2	1	11	10	6	5	6	9	9		
	PrecStd	42	7	5	7	11	15	20	30	29	22	12	10	10		
	DB	-2.3	-1.2	6.8	15.3	27.6	29.8	32.5	28.0	20.3	13.7	4.2	-1.7	-1.7		
	MCWB	-2.8	-2.8	1.8	6.2	13.3	16.1	18.4	16.8	12.8	7.6	2.8	-2.7	-2.7		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	-4.6	-3.6	3.6	12.0	23.9	27.3	29.8	26.0	16.8	10.3	1.6	-4.5	-4.5		
	2% MCWB	-5.1	-4.9	-0.5	4.3	11.9	15.6	18.4	16.8	11.0	5.5	0.6	-5.1	-5.1		
	5% DB	-7.7	-5.7	1.8	8.8	20.5	25.2	28.2	24.4	14.7	8.2	0.2	-6.7	-6.7		
	5% MCWB	-8.3	-6.6	-1.4	3.2	11.2	15.0	17.7	16.0	9.6	5.0	-0.5	-7.2	-7.2		
	10% DB	-10.3	-7.6	0.0	6.2	17.6	23.1	26.4	22.8	12.7	6.0	-1.6	-8.9	-8.9		
	10% MCWB	-10.7	-8.4	-2.6	1.6	9.5	14.3	17.2	15.5	8.8	3.1	-2.5	-9.4	-9.4		
	0.4% WB	-2.8	-2.8	1.9	7.1	14.7	18.1	21.1	19.1	13.2	7.9	3.0	-2.6	-2.6		
	0.4% MCDB	-2.5	-1.7	5.8	13.8	24.7	25.5	27.0	22.2	18.1	11.4	4.5	-1.9	-1.9		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	-5.0	-4.9	0.1	5.0	13.1	17.0	19.8	17.8	12.1	6.5	0.7	-5.0	-5.0		
	2% MCDB	-4.5	-4.0	2.9	11.3	22.0	24.4	25.9	22.9	15.9	9.7	1.3	-4.6	-4.6		
	5% WB	-8.3	-6.6	-1.1	3.4	11.9	16.0	19.0	17.0	10.7	5.1	-0.5	-7.2	-7.2		
	5% MCDB	-7.8	-5.8	1.5	8.1	19.0	22.9	24.8	23.1	13.5	7.8	0.3	-6.8	-6.8		
	10% WB	-10.7	-8.3	-2.5	1.9	10.4	14.9	18.2	16.2	9.4	3.3	-2.4	-9.4	-9.4		
	10% MCDB	-10.2	-7.6	0.0	5.8	16.6	21.5	24.3	21.8	12.0	5.5	-1.7	-9.0	-9.0		
	MDBR	5.6	7.1	8.6	7.7	9.3	9.5	9.4	8.7	7.0	6.1	6.2	6.5	6.5		
	5% DB	6.5	7.7	9.0	9.8	13.4	11.9	11.9	11.1	9.7	8.4	5.8	7.3	7.3		
(36) Mean Daily Temperature Range	MCDBR	6.3	7.1	6.3	6.2	7.0	5.1	4.4	4.4	4.9	5.5	5.4	7.1	7.1		
	MCWBR	6.6	7.3	8.3	9.0	11.7	10.9	10.1	9.3	8.2	7.4	5.7	7.3	7.3		
	MCDBR	6.4	6.9	6.2	6.0	6.6	4.8	4.3	4.3	4.8	5.5	5.3	7.3	7.3		
	MCWBR	6.4	6.9	6.2	6.0	6.6	4.8	4.3	4.3	4.8	5.5	5.3	7.3	7.3		
(40) Clear-Sky Solar Irradiance	taub	0.272	0.301	0.325	0.359	0.385	0.423	0.459	0.411	0.345	0.328	0.283	0.255	0.255		
	taud	1.982	2.007	2.044	2.146	2.272	2.236	2.138	2.249	2.416	2.262	2.042	1.945	1.945		
	Ebn at Noon	526	699	802	843	847	815	770	784	795	680	508	415	415		
	Edh at Noon	59	92	118	127	121	127	138	115	83	72	55	44	44		
(44) All-Sky Solar Radiation	RadAvg	0.64	1.60	3.08	4.45	5.15	5.78	5.36	4.25	2.61	1.26	0.62	0.37	0.37		
	RadStd	0.07	0.12	0.18	0.31	0.39	0.45	0.43	0.27	0.32	0.13	0.07	0.06	0.06		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page